MSAHiFiC: A Super-Prior Driven High-Fidelity Spectral Attention Network for Hyperspectral Image Compression

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***Abstract*—To address the limitations of existing deep learning-based hyperspectral image compression methods in accurately modeling the rate-distortion problem, we propose a coupled multiscale attention spatialspectral high-fidelity compression (MSAHiFiC) network. MSAHiFiC employs a super-prior network to estimate bitrate and guide rate-distortion optimization, enhancing performance under constrained bitrate conditions. A multiscale spectral attention module (MSAM) is introduced to capture spectral dependencies across varying interband distances and preserve key spectral features during downscaling. A spectral fidelity term is further incorporated into the loss function to improve reconstruction accuracy. Experiments on three benchmark hyperspectral datasets-HySpecNet-11k, XiongAn, and WHU-Hi-demonstrate that MSAHiFiC outperforms state-of-the-art methods by achieving 5% higher spectral fidelity and 6% improvement in reconstruction accuracy under a low bitrate of 0.5 bpp.**

***Index Terms*—Compression networks, hyperspectral remote sensing, multiscale attention, rate-distortion optimization, spatial–spectral compression.**

I. Introduction

HYPERSPECTRAL remote sensing technology is developing rapidly, and with its advantage of being able to acquire a detailed spectral response from the target features [1], hyperspectral remote sensing images can realize more delicate tasks. As a result, hyperspectral remote sensing has been successfully applied in many fields, such as land-cover classification and change detection [2], environmental monitoring [3], [4], agriculture [5], forestry identification [6], [7], mineral identification [8], and landslide hazard mapping [9]. However, compared with traditional optical remote sensing or multispectral remote sensing images, hyperspectral remote sensing images have more spectral channels, which means an increase of tens or even hundreds of times in the data volume [10]. Large volumes of hyperspectral image data are difficult to transmit and store due to the limited transmission bandwidth and storage space [11]. Therefore, image compression techniques have attracted the attention of many scholars [10]. Many natural image compression methods have been improved and introduced into the hyperspectral field, giving rise to numerous typical hyperspectral remote sensing image compression methods [12]. These methods have gradually formed two main approaches: 1) model-driven methods and 2) deep learning-based methods.

The model-driven hyperspectral image compression methods explicitly simulate the relationship between the original hyperspectral image and the compressed features in the image compression problem. The compressed features obtained by transforming the original image through a quantizer are discretized, and the quantized data are entropy encoded for lossless compression, forming a data stream that is transmitted or organized into files. These methods include domain transformation models [13], prediction models [14], dictionary models [15], and decomposition models [16]. Moreover, the classic domain transformation models include 3DDCT [17], 3DSPECK [18], and PCA-JPEG2K [19]; predictive models such as ADPCM [20], 3DMBLP [21], and C-SNM-RLS-LMS [22]; dictionary models such as MVQ [23], PWSR [24], and SSASR [25]; and decomposition models such as PSO-NTD [26] and CNN-NTD [27].

The model-driven hyperspectral remote sensing image compression methods usually focus on concise feature representations rather than rate-distortion issues, resulting in their rate-distortion curves being determined after the model is determined. Furthermore, due to the large spatial scale of hyperspectral images and the obvious differences in the

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features of each region, it is difficult for the traditional methods to achieve an optimal rate-distortion performance by using the same model to encode all the regions. The deep learning-based hyperspectral image compression methods overcome the single model problem of the traditional methods by implicitly learning the relationship between the original image and the compressed features through the network. Such methods can be divided into coupled deep learning modeling methods and direct deep learning compression methods.

The coupled deep learning approaches retain the traditional compression framework while using deep neural networks to implicitly learn key compression steps. These models are particularly suitable for integration with conventional methods. For instance, the temporal modeling capability of recurrent neural networks enables effective spectral interprediction in hyperspectral data, facilitating efficient spectral intercompression. Typical methods include C-DPCM-RNN [28] and GRE-DBN [29].

The direct deep learning-based compression methods do not consider an explicit compression model and directly use the encoder to obtain the key features for quantization coding. They then use a decoder to reconstruct the image, and continuously optimize the reconstruction distortion loss to achieve end-to-end rate-distortion optimization. Typical approaches include 1DCAE [30], SSCNet [31], and 3DCAE [32] utilizing a convolutional autoencoder (CAE), and HiFiCSE [33] with a coupled autoencoder and conditional generative adversarial network (cGAN).

Currently, numerous deep learning-based compression methods for hyperspectral remote sensing images adopt network architectures originally designed for natural image compression. However, these approaches often inadequately account for the distinctive feature scales and spectral structures inherent in hyperspectral imagery [34]. Most existing techniques focus primarily on spatial feature extraction and have yet to effectively optimize the spatial–spectral rate-distortion trade-off, resulting in limited performance in achieving high-fidelity reconstruction at low bit rates. The efficient compression of hyperspectral data remains challenging due to significant redundancies across both spectral and spatial dimensions, the necessity to preserve subtle spectral information, and the critical balance required between compression ratio and reconstruction quality. These persistent issues continue to complicate the development of robust and efficient compression algorithms. Addressing these challenges is essential not only for enhancing compression efficiency but also for enabling effective transmission, storage, and subsequent analysis of hyperspectral data, particularly in large-scale and real-time application scenarios.

In this article, we propose a novel deep learning-based method for hyperspectral image compression that integrates a self-encoder network with a super-prior mechanism to adaptively optimize the code rate. Unlike existing approaches that often rely on handcrafted priors or generic architectures tailored for natural images, our framework incorporates a multiscale spectral attention module (MSAM) designed to capture both local and global spectral dependencies effectively, thereby significantly enhancing rate-distortion performance. Furthermore, we introduce a dedicated rate-distortion constraint network that not only refines the loss function but also systematically explores the influence of spatial–spectral priors—a dimension largely underutilized in current methods. These innovations collectively improve reconstruction fidelity at low bit rates while substantially alleviating storage and transmission demands. The main contributions of the proposed method are as follows.

1) *Enhanced Super-Prior Network for Joint Probability Estimation*: Unlike conventional autoencoder-based bottlenecks that rely on fixed priors, we introduce a super-prior network that explicitly models the mean and variance of the compressed features. This approach enables joint probability estimation and achieves more accurate and adaptive bitrate prediction, significantly improving rate-distortion optimization compared to existing methods.
2) *MSAM for Adaptive Correlation Modeling*: We design a novel multiscale spectral attention mechanism that captures both local and global spectral dependencies through varied receptive fields. This allows the model to adaptively emphasize informative spectral bands and preserve critical spectral details—addressing a common limitation in existing approaches that treat spectral regions uniformly or use limited contextual information.
3) *Integrated Rate-Distortion Optimization With Spatial–Spectral Constraints*: Moving beyond traditional loss functions that often prioritize spatial reconstruction, we propose a comprehensive optimization objective incorporating rate estimation, spectral–spatial distortion metrics (e.g., mse and SAM), and adversarial training. This holistic loss framework effectively balances compression rate with multidimensional reconstruction quality, enhancing fidelity in both spatial and spectral domains under low-bitrate scenarios.

In this study, experiments were conducted on the HySpecNet-11k [35], XiongAn [36], and WHU-Hi [37] datasets, and the results were compared with those of the traditional hyperspectral image compression methods and deep learning-based methods. The experimental results confirm that the proposed method can achieve superior rate-distortion optimized compression, compared to the other methods.

The rest of this article is organized as follows. Section II briefly introduces the CAE and GAN and their compression methods. Section III describes the proposed multiscale attention spatial–spectral high-fidelity compression (MSAHiFiC) compression method in detail. The experimental results are provided in Section IV. This article concludes with a summary and discussion in Section V.

II. Related Research Background

Most of the deep learning-based hyperspectral image compression methods are currently implemented based on a CAE, which reduces the dimensionality of the potential space by sequentially applying convolution (conv) and downsampling operations [38]. However, since CAE-based methods have

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fixed compression ratios and relatively high bit rates, image compression models based on GANs have been proposed [39], which are capable of achieving higher compression ratios and ensuring generated images with better quality.

A. CAE and cGANs

The CAE is an unsupervised learning-based neural network model that consists of an encoder and a decoder. The encoder compresses the input high-dimensional data into low-dimensional feature representations, and the decoder restores these feature representations to the original data. The goal of an autoencoder is to minimize the difference between the encoder and the decoder [40].

The GAN is a generative model widely used in the field of natural image compression in recent years. Its core idea is derived from the binary zero-sum game of game theory [41], which regards the generative problem as an adversarial game between a generator and a discriminator. The generator learns to generate samples that are close to the true data distribution from a given noise distribution, and the generated instances become negative training examples for the discriminator, while the discriminator learns to discriminate whether the input data are from the real data or the generator [42]. The cGAN introduces conditional information that effectively compensates for the uncertainty of the GAN [43]. In the cGAN-based methods, the features extracted by the encoder are used as conditional inputs, which enable the generator to reconstruct a high-quality compressed image. Meanwhile, the discriminator is used to evaluate the compressed image and the original image. The two networks promote each other, and the samples generated by the generator get closer and closer to the real samples. Image compression based on a cGAN ensures that the compressed image is of good quality and has a high compression ratio. The details are mentioned as follows.

1) The original image is encoded into a low-dimensional representation using an encoder.
2) The low-dimensional representation is decoded into an image using a decoder.
3) The decoded image is evaluated using a discriminator to calculate its similarity with the original image.
4) Two loss functions are used to optimize the generator and the discriminator, respectively, so that the compressed image is close to the original image.

B. Channel Attention-Based High-Fidelity Compression (HiFiCSE)

Byju et al. [33] proposed channel attention-based high-fidelity compression (HiFiCSE), which consists of four modules: 1) encoder E; 2) probability model P; 3) generator G; and 4) discriminator D. The input hyperspectral image X is encoded into spatially and spectrally decorrelated latent representations Y by the encoder E and quantizer Q. It is then entropy encoded into a bit stream using arithmetic coding with a minimum bit rate $r(y_i) = -\log(P(y_i))$ to decode the bit stream further back to Y. The generator decodes back to X', finally, the discriminator D with conditional information (Y) distinguishes whether the reconstructed image is real or generated.

Each module E, P, G, and D of the method is parameterized by a convolutional neural network and optimized together to obtain a minimum rate-distortion tradeoff with a loss function, as shown in (1) and (2), where $r(Y)$ is the rate, $d(X, X')$ is the distortion loss, λ is a hyperparameter controlling the rate-distortion tradeoff, $\log(D(X', Y))$ is the conditional discriminator loss, and β controls the discriminator loss effect. The distortion loss is shown in (3), where θ is the hyperparameter controlling the effect of the mse, structural similarity index (SSIM), and learned perceptual image block similarity (LPIPS) loss in the total distortion loss calculation

$$L_{E,G,P} = \mathbb{E}_{X\sim P_X}\left[\lambda r(Y) + d\left(X, X'\right) - \beta \log D\left(X', y\right)\right] \quad (1)$$

$$L_D = \mathbb{E}_{X\sim P_X}\left[-\log\left(1 - D\left(X', Y\right)\right)\right] + \mathbb{E}_{X\sim P_X}\left[-\log D\left(X, Y\right)\right] \quad (2)$$

$$d\left(X, X'\right) = \theta_1 \cdot \text{mse} + \theta_2 \cdot (1 - \text{SSIM}) + \theta_3 \cdot \text{LPIPS}. \quad (3)$$

The HiFiCSE method improves the rate-distortion performance of hyperspectral image compression to some extent. However, it overlooks the complex spatial–spectral features of hyperspectral remote sensing images and pays insufficient attention to the overall structure of spectral vectors, which can result in significant spectral distortion.

Algorithm 1 Pseudo-Code for MSAHiFiC

Input: $\mathbf{x} \in \mathbb{R}^{\mathrm{B}\times\mathrm{C}\times\mathrm{H}\times\mathrm{W}}$

for $i = 1, 2, \ldots, N_{\text{epochs}} \in 150$
  $x \leftarrow MSAM(x)$
  $y \leftarrow$ Encoder (x)
  $\mathrm{z} = \mathrm{P}(y)$
  $\hat{\mathbf{y}} = \text{EntropyEncode}(\hat{\mathbf{y}}, \hat{\mu}, \hat{\sigma})$
  $\hat{\mathbf{x}} = \text{Decoder}(\hat{\mathbf{y}})$
  $\text{Mind}(\hat{\mathbf{x}}, \mathbf{x}) = \lambda_1 \text{mse} + \lambda_2 \text{SAM}$
End
Update Encoder, Decoder, Super-Prior
for $i = 151, 152, \ldots, N_{\text{epochs}} \in 150$
  $\hat{\mathbf{x}} =$ Generator $(\hat{\mathbf{y}})$
  $D_{\text{fake}} = D(\hat{\mathbf{x}}, \mathbf{y})$
  $r(\hat{\mathbf{y}}, \hat{\mathbf{z}}) = -\log P(\hat{\mathbf{y}}, \hat{\mathbf{z}} \mid \hat{\mu}, \hat{\sigma})$
  $L_{E,G,H} = E_{x-p_x}[r(\hat{y}, \hat{z}) + d(\hat{x}, x) - \beta \log D(\hat{x}, y)]$
  $L_D = E_{x-p_x}[-\log(1 - D(\hat{x}, y)) - \log D(x, y)]$
end
Update $\theta_D \leftarrow \theta_D - \nabla_{D;}\theta_{E,G,H} \leftarrow \theta_{E,G,H} - \nabla_{E,G,H}$

III. Proposed MSAHiFiC Method

The overall framework of the proposed MSAHiFiC method is shown in Fig. 1, the pseudo-code is provided in Algorithm 1, and the detailed process is presented as follows. The proposed method consists of four main parts: encoder, super-prior, generator, and discriminator. The encoder combines the MSAM to downscale the original image x using convolutional pooling to obtain the compressed feature y. The encoder is used to generate the compressed feature y by using convolutional pooling. The compressed feature is

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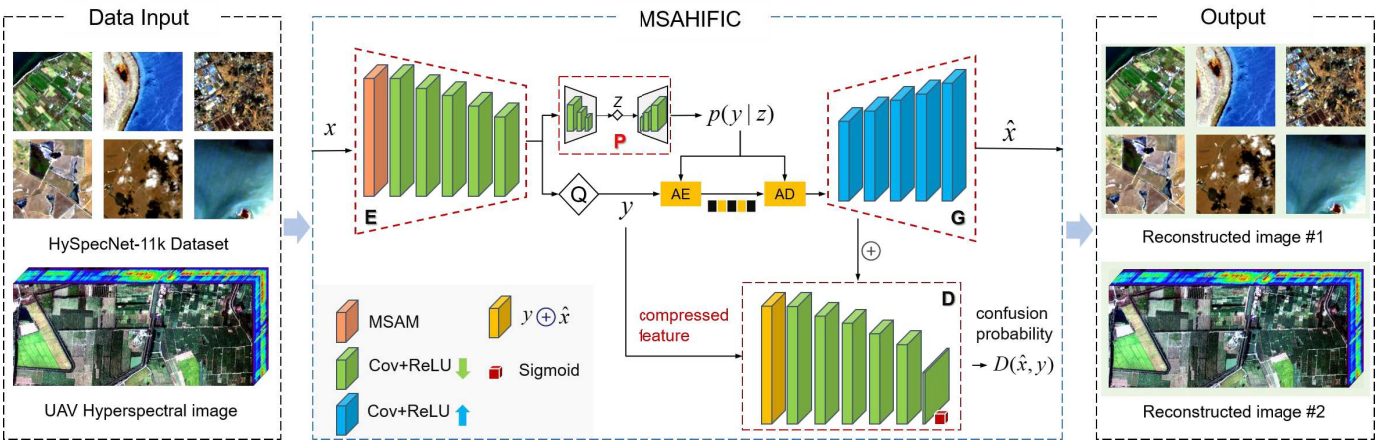


Fig. 1. Network structure of MSAHiFiC. The proposed method consists of four main parts: (E) encoder, (P) super-prior, (G) generator, and (D) discriminator.

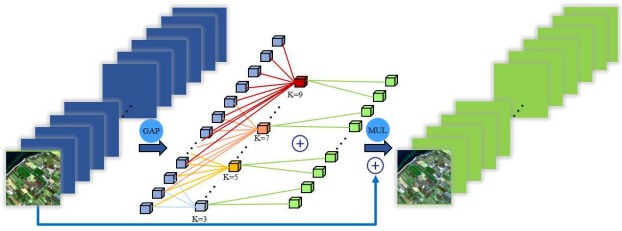


Fig. 2. Multiscale spectral attention module.

extracted from its super-prior z by the analyzer network of the prioritizer, and the distributional information $P(Y|Z)$ is generated using the synthesis network, thus serving as a priori information to guide the lossless coding process of the quantized feature. The generator outputs the reconstructed image $\hat{x}$ through the structure of decoding feature optimization and encoder mirroring. The discriminator draws on the idea of the cGAN, splicing the auxiliary conditional information x with the reconstructed image $\hat{x}$ to generate the probability of discriminatory confusion, thus optimizing the generator and the discriminator.

A. Design of the MSAM

The spectral dependence of hyperspectral remote sensing images exhibits a solid nonlinear structure, with band correlations not simply decreasing with distance, and some bands spanning a large range being highly correlated. However, most of the traditional convolution kernel sizes are small, limiting their receptive fields to extract only a narrow range of correlation information, which hinders the extraction of the long-range spectral dependencies [44].

To extract this nonlinear dependency [45], the MSAM is introduced in the proposed network. As shown in Fig. 2, the pseudo-code is provided in Algorithm 2. The specific design includes the following key points.

1) *Multiscale Convolution:* Use different sizes of convolution kernels (e.g., 3×3, 5×5, 7×7, and 9×9) to capture the spectral dependencies at different scales.
2) *Feature Splicing and Fusion:* The output of multiscale convolution is spliced in the spectral dimension and fused by 1×1 convolution to generate spectral attention maps.

Algorithm 2 Pseudo-Code for MSAM

Input: $\mathbf{x} \in \mathbb{R}^{\mathrm{B}\times\mathrm{C}\times\mathrm{H}\times\mathrm{W}}$
Output: $\boldsymbol{F_0} \in \mathbb{R}^{B\times H\times W}$
identity $\leftarrow \boldsymbol{x}$
$\boldsymbol{x} \leftarrow$ Unsqueeze $(\boldsymbol{x}, \mathrm{dim} = 1)$
avg $\leftarrow$ GlobalAvgPool$(\boldsymbol{x})$
$\mathrm{res_i}$ = Conv3D(avg, kernel = 3, 5, 7, 9)
res $\leftarrow$ Concat(re $\pm$,...,res4, dim=C)
res $\leftarrow$ Squeeze(res, dim=1)
attn = σ(Conv2D(res, kernel=1))
$\boldsymbol{x} \leftarrow \boldsymbol{x} \times$ **attn**
$\boldsymbol{x} \leftarrow$ Conv2D(Norm(ReLU(Conv2D($\boldsymbol{x}$))))
$F_o \in \mathbb{R}^{B\times H\times W} \leftarrow x+$ identity

3) *Dynamic Weight Adjustment:* The attention map is generated by a Sigmoid activation function and multiplied element by element with the input feature map to dynamically adjust the importance of each band.

For the input feature map $F_I \in \mathbb{R}^{B\times H\times W}$, a global average pooling (GAP) operation is used to aggregate the spatial information of each spectral segment, to generate feature vectors $S \in \mathbb{R}^B$, as shown in the following equation:

$$S(b) = \mathrm{GAP}(F_I(b)) = \frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W} F_I(b,h,w). \quad (4)$$

In addition, the first plus sign in Fig. 2 represents the concatenation of multiscale convolution results along the channel dimension, and the second plus sign represents the residual connection, where the reweighted features are added to the original input. The "MUL" operation above the second plus sign represents the elementwise multiplication of the original input and the generated attention weights, achieving feature reweighting to enhance important features while suppressing less relevant ones. For the aggregated spatial information, 3-D convolution operations with different convolution kernel sizes are used to capture the spectral dependencies at different

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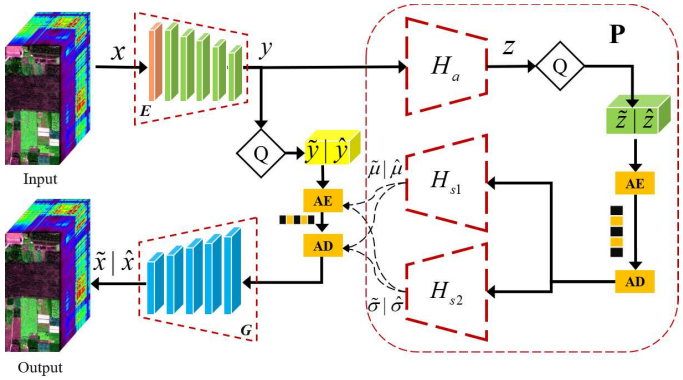


Fig. 3. Super-prior network architecture.

scales. Considering that each spectral segment can be associated with a different number of neighboring spectral segments at different distances, these captured dependencies are spliced in the spectral dimension for the subsequent information fusion. In (5), W_k refers to the convolution weights with a convolution kernel size of $k \times 1 \times 1$. A 1×1 convolution is then utilized to fuse these spectral dependencies. Finally, a sigmoid activation function is utilized to generate the spectral attention maps, and the output feature maps are formed by $F_I \in \mathbb{R}^{B\times H\times W}$ weighting the input feature maps $F_o \in \mathbb{R}^{B\times H\times W}$

$$S_d = \text{concat}\left(W_3 * S, W_5 * S, W_7 * S, W_9 * S\right) \tag{5}$$

$$F_O = \text{Sigmoid}\left(W_f * S_d\right) \otimes F_I \tag{6}$$

where W_f is the convolution weight used in the fusion process. It should be noted that the MSAM only utilizes the interspectral dependencies to weigh each spectral feature. Therefore, it needs to be embedded into the residual block, which first acquires the multiscale spectral attention map, based on which it is processed using two layers of 3×3 convolution, normalization, and activation, and the residuals are connected to construct the output feature map.

B. Variational Super-Prior Networks

The self-encoder network combined with a cGAN network effectively extracts compression features but does not optimize the rate-distortion performance. According to Shannon's information theory [46], the code rate of compressed features depends on their information entropy, which is influenced by their probability distribution. Studies have indicated that the downscaled features from the encoder still contain some spatial and spectral redundancy and have nonuniform probability distributions [47]. To address this issue, the proposed method introduces a variational super-prior network. This network learns the probability distribution prior—essentially the "prior of the prior"—for the hidden compression features, enhancing the rate-distortion performance by better modeling the probability distribution.

The variational super-prior network builds on the principles of the variational autoencoder (VAE). The VAE addresses the issue of limited generalization in autoencoder networks by modeling the input sample space using a Gaussian mixture model. This approach assumes that each input sample is distributed according to a Gaussian distribution $x_i \sim \mathbb{N}(\mu_i, \sigma_i^2)$, which helps improve the network's generalization capabilities. The encoder is utilized to obtain this distribution information, coupling noise that satisfies a zero-mean Gaussian distribution into the decoder to produce an output with the same distribution. The specific design includes three key points: latent variable modeling, entropy coding optimization, and joint training.

TABLE I
PRIOR MODEL NETWORK ARCHITECTURE

Network	L	K	S	Input	Output	AF
H_a	Conv4-1	3	1	C	320	ReLU
	Conv4-2	5	2	320	320	ReLU
	Conv4-3	5	2	320	320	-
H_s	TrConv4-4	5	2	320	320	ReLU
	TrConv4-5	5	2	320	320	ReLU
	TrConv4-6	3	1	320	C	-

Notes: H_a : prior analysis model, H_s : prior synthesis model, L: layer, K: kernel size, S: stride, AF: activation function.

The variational super-prior network in the proposed method comprises three main components, as illustrated in Fig. 3: an analytical model H_a and two integrated models H_{s1} and H_{s2}. The network structure is detailed in Table I. The analytical model encodes the downsampled latent feature y into a higher level super-prior z, which guides the entropy model in accurate bitrate estimation. The two integrated models then decode this super-prior z to compute the mean and variance of the compression feature y, enabling precise probability modeling and improved reconstruction. This design allows the network to adaptively capture the intrinsic spatial and spectral redundancies in hyperspectral images, providing a more informative prior to enhance rate-distortion optimization. To efficiently estimate the code rate and optimize the rate-distortion performance, the network employs a univariate nonparametric multilayer density model for the super-prior z. This model characterizes the cumulative probability distribution of each channel variable $\boldsymbol{c}$ using (7) and maps it to the range of 0–1 through a sigmoid operation

$$f_k(\boldsymbol{c}) = \begin{cases} g_k\left(\boldsymbol{H}^{(k)}\boldsymbol{c} + \boldsymbol{b}^{(k)}\right), & 1 \le k < K \\ \text{sigmoid}\left(\boldsymbol{H}^{(k)}\boldsymbol{c} + \boldsymbol{b}^{(k)}\right), & \text{otherwise} \end{cases} \tag{7}$$

where k represents the number of layers of the density model; $\mathbf{H}^{(k)}$ and $\boldsymbol{H}^{(k)}$ are, respectively, the weight and bias parameters of the kth layer; and g_k is the function that introduces the nonlinearity, which is defined as shown in the following equation:

$$g_k(\boldsymbol{c}) = \boldsymbol{c} + \boldsymbol{a}^{(k)} \odot \tanh(\boldsymbol{c}). \tag{8}$$

In (8), $\boldsymbol{a}^{(k)}$ is the size parameter of the kth layer, and its value controls the expansion and contraction of the density function. $\odot$ represents the element-by-element product, and when the scale parameter $\boldsymbol{a}^{(k)}$ is set to a positive value, the peak of the density function will be more significant. In addition, considering the differential nonnegativity of the cumulative probability distribution, which may not be satisfied by the actual parameters, the weight parameter $\mathbf{H}^{(k)}$ is

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constrained using the soft-plus function, while parameter $\boldsymbol{a}^{(k)}$ is reparametrized by tanh, as shown in the following equations:

$$\mathbf{H}^{(k)} = \text{softplus}\left(\hat{\mathbf{H}}^{(k)}\right) \tag{9}$$

$$\boldsymbol{a}^{(k)} = \tanh\left(\widehat{\boldsymbol{a}}^{(k)}\right). \tag{10}$$

All parameters are tuned for network optimization to ensure that the univariate nonparametric multilayer density model accurately captures the cumulative probability distribution of the super-prior z. Study has confirmed that the probability density from this model closely aligns with the probability distribution of z derived from a Gaussian mixture model, enabling precise estimation of the code rate for the super-prior z [47].

For any variable y_i in the compression feature y, since it is modeled as a Gaussian distribution $y_i \sim \mathbb{N}(\mu_i, \sigma_i^2)$, its mean and variance can be determined using the integrated model of the prior. This allows for an accurate estimation of its coding rate, as described in the following equation:

$$R = \mathbb{E}_{\hat{y}}\left[-\log_2 p_{\hat{y}}(\hat{y})\right]. \tag{11}$$

C. Network Loss Function Construction

The core of the hyperspectral image compression network is a self-encoder structure with a super-prior, as illustrated in Fig. 3. The network's loss function is designed based on the compression feature y and the super-prior z.

Given input hyperspectral data, x the network input target of the encoder is the joint probability posterior distribution of y and z, which can be expressed as $p_{\tilde{y},\tilde{z}|x}(\tilde{y},\tilde{z}|x)$ after quantization (utilizing uniform noise substitution for optimization). However, the actual output distribution of the network can be set to $q_{\tilde{y},\tilde{z}|x}(\tilde{y},\tilde{z}|x)$. To make these two distributions as similar as possible, their distances are modeled using the Kullback–Leibler scatter, as shown in (12). The multivariate Bayesian formulation can then be used to derive (13)

$$\begin{aligned}&\mathbb{E}_{\mathcal{X}\sim p_x} D_{\mathrm{KL}}\left[q_{\tilde{y},\tilde{z}|x}\|p_{\tilde{y},\tilde{z}|x}\right]\\&= \mathbb{E}_{\mathcal{X}\sim p_x}\mathbb{E}_{\tilde{y},\tilde{z}\sim q}\left[\log q(\tilde{y},\tilde{z}|x) - \log(\tilde{y},\tilde{z}|x)\right]\end{aligned} \tag{12}$$

$$\begin{aligned}&p(x,\tilde{y},\tilde{z})\\&= p(\tilde{y},\tilde{z}\mid x)\,p(x) = p(x\mid\tilde{y},\tilde{z})\,p(\tilde{y}\mid\tilde{z})\,p(\tilde{z})\end{aligned} \tag{13}$$

$$\begin{aligned}&\mathbb{E}_{x\sim p_x} D_{KL}\left[q_{\tilde{y},\tilde{z}|x}\|p_{\tilde{y},\tilde{z}|x}\right]\\&= \mathbb{E}_{x\sim p_x}\mathbb{E}_{\tilde{y},\tilde{z}\sim q}\left[\log q(\tilde{y},\tilde{z}\mid x) - \log p(x\mid\tilde{y},\tilde{z})\right.\\&\qquad\left. - \log p(\tilde{y}\mid\tilde{z}) - \log p(\tilde{z})\right] + \text{const.}\end{aligned} \tag{14}$$

At the same time, in (14), the term "const" denotes the logarithm of the prior probability of the input sample x, that is, $\log p(x)$. Since x is fixed during the optimization process, this term remains constant and has no impact on the gradient-based learning. As such, it is typically omitted from the loss function during training and retained in the derivation solely for mathematical completeness. Considering that $q_{\tilde{y},\tilde{z}|x}(\tilde{y},\tilde{z}|x)$ is computed using a uniform noise of two unit widths for its processing, the first term will be approximated to 0. For the second term, it is equivalent to the reconstruction error that arises when reconstructing x using $\tilde{y}$ and $\tilde{z}$. The latter two terms correspond to the cross-entropy between $\tilde{y}$ and $\tilde{z}$, essentially characterizing the code rate constraints in the rate-distortion optimization problem. Of these terms, the fourth term has been optimized using the univariate nonparametric multilayer density model described in Section III-B. Overall, optimization using this loss function is effectively rate-distortion optimization of the network.

TABLE II
SPECIFICS OF THE HSI SCENE CLASSIFICATION DATASETS

Dataset	HySpecNet-11k	XiongAn	WHU-Hi-HanChuan	WHU-Hi-HongHu
Sb	224	256	274	270
Pb	202	256	256	256
Sr/m	30	0.5	0.109	0.043
Size/pixels	128×128	3750×1580	1217×303	940×475
Num	11483	1	1	1

Notes: Sb: number of spectral bands, Pb: number of processed bands, Sr/m: spatial resolution (m), Num: number of images. Download links as follows:
HySpecNet-11k: https://hyspecnet.rsim.berlin/
XiongAn: http://www.hrs-cas.com/a/share/shujuchanpin/2019/0501/1049.html
WHU-Hi: http://rsidea.whu.edu.cn/resource_WHUHi_sharing.htm

In addition, based on the self-encoder structure with super-prior, the proposed method combines the loss function of the cGAN to further optimize the rate-distortion performance of the network. Therefore, the loss function, as shown in (15) and (16), is used in the actual optimization process

$$\mathbb{L}_{E,G,H} = \mathbb{E}_{x\sim p_x}\left[r(\tilde{y},\tilde{z}) + d(\tilde{x},x) - \beta\log D(\tilde{x},y)\right] \tag{15}$$

$$\mathbb{L}_D = \mathbb{E}_{x\sim p_x}\left[-\log(1 - D(\tilde{x},y)) - \log D(x,y)\right] \tag{16}$$

where $r(\tilde{y},\tilde{z})$ is the code rate consumed by the coded compression feature $\tilde{y}$ and super-prior $\tilde{z}$. $d(\tilde{x},x)$ is the distortion error between the reconstructed hyperspectral image and the original image. As shown in (17), the loss function is designed to enable the discriminator D to effectively distinguish between real and reconstructed images, encouraging the generation of reconstruction results that are closer to the real data distribution. The first term $-\log(1 - D(\tilde{x},y))$ forces D to recognize distorted samples, while the second term $-\log D(x,y)$ ensures that the real image can be correctly classified. Through adversarial training, we improve the nonlinear representation of the compressive model, which allows the reconstructed image to maintain higher fidelity in both spectral and spatial dimensions. The mse and SAM are introduced in this term to effectively measure the spatial–spectral distortion between the two. The mse loss describes the overall reconstruction error of the image, while the SAM loss focuses on constraining the spectral fidelity. λ_1 and λ_2 denote the weight parameter

$$d(\tilde{x},x) = \lambda_1\text{mse} + \lambda_2\text{SAM}. \tag{17}$$

IV. EXPERIMENTS AND ANALYSIS

In order to evaluate the effectiveness of the proposed method, it was compared with the traditional JPEG2K method and three typical deep learning-based methods for hyperspectral image compression: SSCNet [31], 3DCAE [32], and HiFiCSE [33].

A. Experimental Data and Parameter Settings

The following hyperspectral image scene classification datasets were utilized in this study: HySpecNet-11k [35]

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Fig. 4. Several typical scenarios for the HySpecNet-11k dataset.

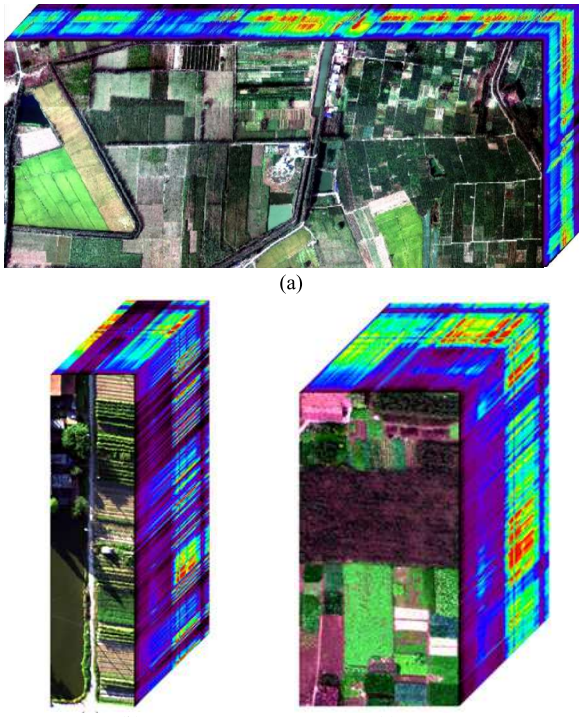


Fig. 5. XiongAn and WHU-Hi hyperspectral image datasets. (a) XiongAn. (b) WHU-Hi-HanChuan. (c) WHU-Hi-HongHu.

and the XiongAn [36], WHU-Hi-HanChuan, and WHU-Hi-HongHu datasets shared by the Intelligent Data Extraction and Analysis of Remote Sensing (RSIDEA) Group of Wuhan University in China [44]. In addition, the XiongAn and WHU-Hi datasets were both cut into 128 × 128 image patches, and the number of bands was downsampled to 256, thus forming an unmanned aerial vehicle (UAV) borne hyperspectral with high spatial resolution (H2) datasets consisting of 387 nonoverlapping image patches. All the datasets were divided into training, validation, and test sets at a ratio of 7:2:1. It is noteworthy that since the UAV datasets are essentially composed of four mixed data groups, stratified sampling must be performed—that is, each of the four data groups should be proportionally partitioned—to avoid compression distortion caused by sensor variations. The specific situation is shown in Figs. 4 and 5 and Table II.

TABLE III
PARAMETER SETTINGS FOR THE HSI SCENE CLASSIFICATION DATASETS

Model	LR	BS	TE1	TE2	LH
SSCNet	1×10^{-4}	1	200	100	-
3DCAE	1×10^{-4}	1	200	100	-
HiFiCSE	1×10^{-5}	1	200	100	λ_1=0.075×2^{-4} λ_2=0.15×2^{-3}, β=0.15
MSAHiFiC	1×10^{-5}	1	200	100	λ_1=0.075×2^{-4} λ_2=0.15×2^{-3}, β=0.15

Notes: LR: learning rate, BS: batch size, TE1: training epochs (HySpecNet-11k), TE2: training epochs (WHU-Hi), LH: λ_1: MSE loss hyperparameter, λ_2: SAM loss hyperparameter, β: hyperparameter weight.

The Adam optimizer was used for training all the deep learning models. The specific parameter settings are listed in Table III. The learning rate (LR) was selected through grid search during preliminary experiments to ensure that the model could reach a relatively stable convergence within 100 epochs. The hyperparameters λ_1 and λ_2, which control the weights of the mse and SAM loss components, respectively, were set based on empirical configurations used in HiFiCSE. The parameter β is used to balance the perceptual loss against the rate-distortion loss, and its value has a noticeable effect on perceived image quality. All models were trained with the same number of epochs (TE1/TE2) and batch size (BS = 1) to ensure fair comparison. For MSAHiFiC and HiFiCSE, a smaller LR of 1×10^{-5} was adopted due to the adversarial training with GANs, which require greater stability to prevent training oscillation. For the qualitative evaluation, a false-color composite image of the reconstructed image was created to assess the overall distortion, and the average spectral curves of typical features were compared to examine the spectral fidelity. For the quantitative evaluation, the peak signal-to-noise ratio (PSNR), SAM, and multiscale structural similarity index (MSSIM) of the reconstructed images were calculated at the same bit rate, and corresponding rate-distortion curves were plotted. In addition, the average encoding time t_E for each image block was used to assess the compression speed, and the model's parameter quantity and computational complexity were also examined.

B. Experiment With the HySpecNet-11k Dataset

In this section, we describe the experimental analysis carried out on the publicly available HySpecNet-11k hyperspectral dataset. We combined qualitative false-color composites of the reconstructed images with the quantitative distortion parameter metrics to evaluate the proposed MSAHiFiC approach. Fig. 6 shows the false-color composites of the reconstructed images and the false-color maps of the original images for various scenes, including farmland, riverbank, town, and ocean, at a 1 bpppb bit rate (compression ratio of 32) for each method.

From the experimental results, it can be found that, under low bit rate conditions, JPEG2K produces obvious block effects due to the downscaled chunking operation of the wavelet transform, and it is difficult to distinguish specific

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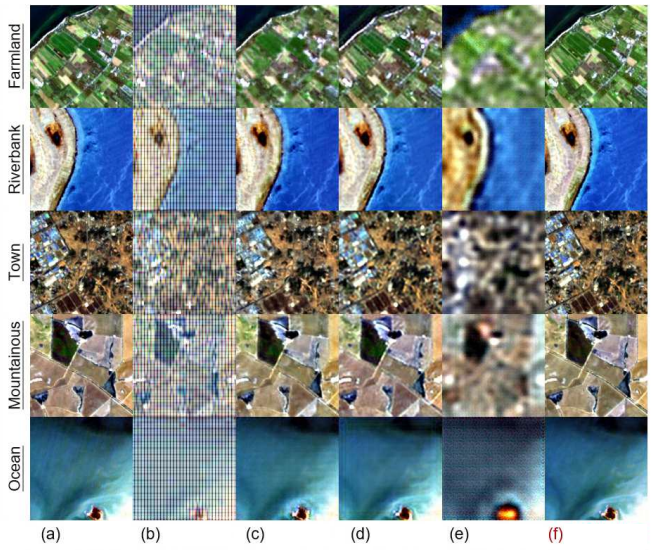


Fig. 6. False-color composites of the reconstructed images of the various methods on the HySpecNet-11k dataset.

TABLE IV
RATE-DISTORTION METRIC EVALUATION FOR THE HYSPECNET-11K DATASET AT1 BPPPB

Metric	JPEG2K	SSCNet	3DCAE	HiFiCSE	MSAHiFiC
SAM ⇩	17.825	3.131	5.798	8.093	**2.837**
PSNR ⇧	23.156	43.913	39.065	34.194	**43.953**
MSSIM ⇧	0.667	0.970	0.935	0.826	**0.974**

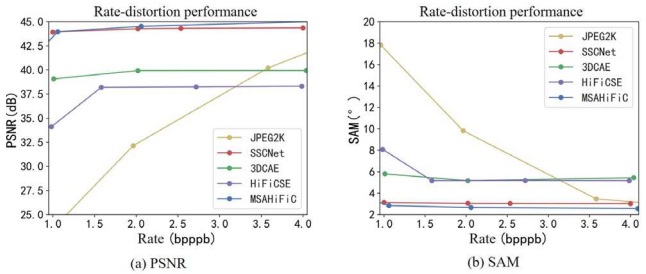


Fig. 7. Rate-distortion curves on the HySpecNet-11k dataset. (a) PSNR. (b) SAM.

features visually. Meanwhile, within each block of JPEG2K, the pixels are highly smoothed and similar, and the edge information is lost to a large degree, but the overall brightness is improved. This is mainly because JPEG2000 employs wavelet-based block partitioning, where each block is encoded independently. At low bitrates, high-frequency components are heavily quantized, and interblock continuity is not well preserved, resulting in distinct blocking artifacts and over-smoothed textures. HiFiCSE has more downscaling layers, which leads to the reconstructed image showing obvious blurring. Furthermore, many farmland features are connected into blocks, and the distortion of the color features is more serious, as can be seen in the riverbank scene. For the town scene, HiFiCSE can no longer identify the specific distribution of the buildings; the cloudy mountain scene has a serious loss of color contrast, and the lower islands of the ocean scene are blurred into bright red patches. The reconstruction results of SSCNet show the loss of many bright feature colors, especially the various colored roofs in the town and the different colored plots in the mountain scene. This is because SSCNet does not consider the extraction of spectral dimension correlations during the compression process, which results in severe spectral distortion. Meanwhile, in the ocean scene, the SSCNet result shows obvious tessellated splicing distortion, and loses the irregular wave texture that exists in the image itself. In comparison, the color details of 3DCAE are richer, and the overall texture is clearer, but the blurring distortion of the mountainous areas in the town scene is more severe. In addition, the edges of the landmasses in the mountain scene are blurred, the detailed features of the cloud thickness are lost, and the wave features in the ocean area are weakened. For 3DCAE, although the 3-D convolutional architecture captures spatial–spectral features, its deep downsampling layers tend to excessively compress information. This leads to the loss of fine-grained textures and boundaries, especially in regions with complex structures, manifesting as blurring and coarse block-level distortions in the reconstruction. The proposed MSAHiFiC method results in richer overall texture details, greater restored color contrast, and the highest similarity with the original image in the various scenes. This fully verifies the advantages of the proposed MSAHiFiC method in preserving the spatial–spectral details of satellite-borne hyperspectral remote sensing images.

Table IV lists the rate-distortion metric scores for compression at a rate of 1 bpppb (compression ratio of 32) on the HySpecNet-11k dataset. Among the different methods, JPEG2K has the lowest values for the SAM, PSNR, and MSSIM metrics, which is consistent with the visual results. HiFiCSE achieves an average structural similarity of 0.826, indicating that the overall structural distribution of the reconstructed image is similar to that of the original image. However, the SAM and PSNR remain at lower levels, corresponding to a significant loss of texture and color details. The SSCNet and 3DCAE autoencoder structures perform relatively well overall. Meanwhile, MSAHiFiC, which incorporates SSCNet's network structure and couples it with the MSAM, shows improvements across all the metrics. The SAM, in particular, is reduced by 0.3°, indicating the higher spectral fidelity of the network.

To illustrate the relationship between distortion and bit rate for the various hyperspectral compression methods, Fig. 7 shows the rate-distortion curves for the PSNR and SAM metrics on the HySpecNet-11k dataset. It can be observed that JPEG2K exhibits a significant variation in performance with changes in bit rate, with a sharp decline in quality at lower bit rates. In contrast, HiFiCSE and 3DCAE show more stable rate-distortion curves, with the PSNR fluctuating slightly around 38 and 40 dB, while the SAM remains around 5°. The proposed MSAHiFiC method, which has a basic codec structure that is similar to that of SSCNet, benefits from the introduction of the spectral attention module and the synergistic optimization with GAN loss. Consequently, its PSNR and SAM variation curves are consistently above those of SSCNet, and it achieves the best rate-distortion performance.

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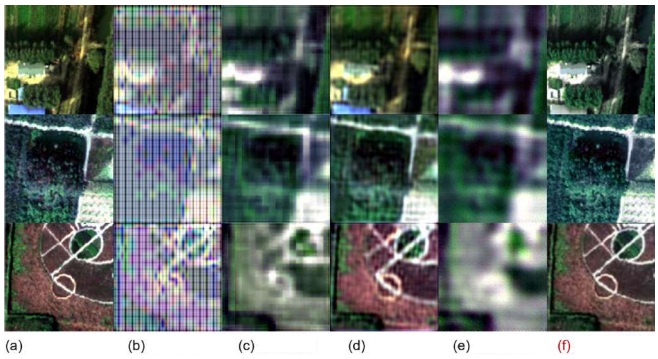


Fig. 8. Reconstructed image blocks of different methods on the XiongAn dataset. (a) Original image. (b) JPEG2K. (c) SSCNet. (d) 3DCAE. (e) HiFiCSE. (f) MSAHiFiC.

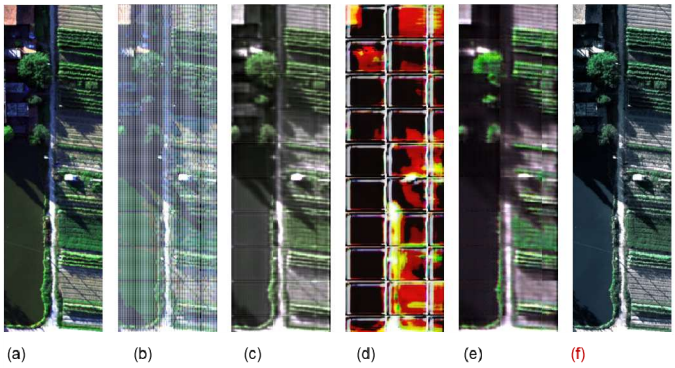


Fig. 9. Reconstructed image blocks of the different methods on the WHU-Hi-HanChuan dataset. (a) Original image. (b) JPEG2K. (c) SSCNet. (d) 3DCAE. (e) HiFiCSE. (f) MSAHiFiC.

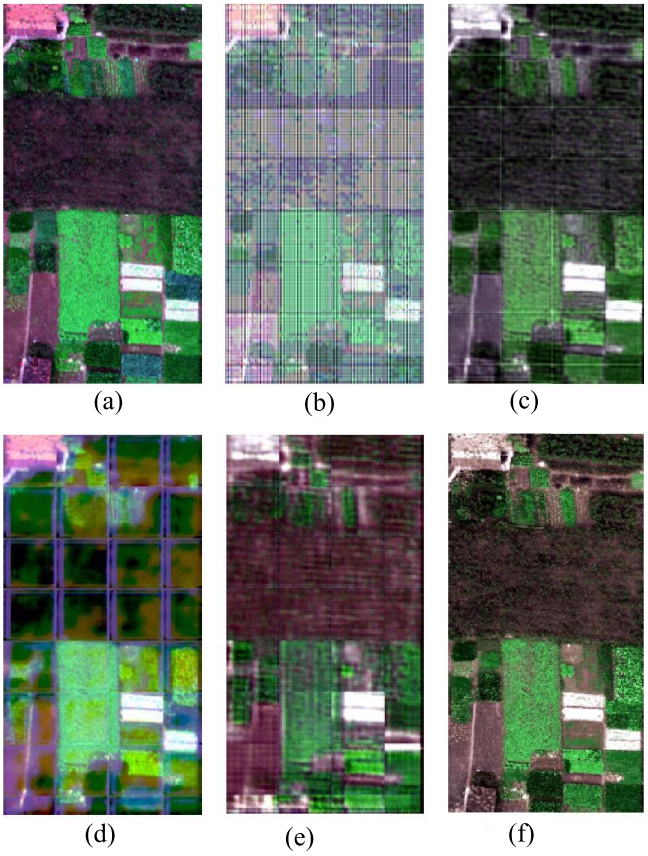


Fig. 10. Reconstructed image blocks of the different methods on the WHU-Hi-HongHu dataset. (a) Original image. (b) JPEG2K. (c) SSCNet. (d) 3DCAE. (e) HiFiCSE. (f) MSAHiFiC.

C. Experiment With the Whu-Hi Dataset

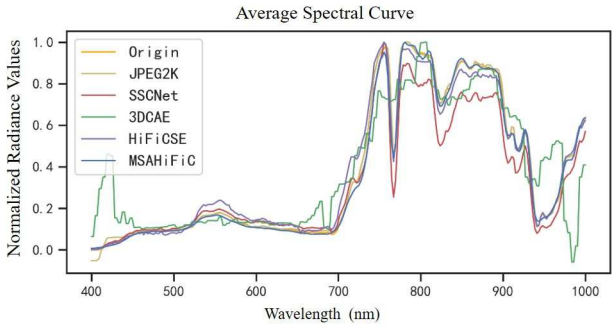
In this section, we describe the training and testing of the compression networks carried out on the WHU-Hi dataset. The reconstructed image maps and spectral maps of the different methods are provided to compare the spatial–spectral rate-distortion parameter metrics, and the rate-distortion curves are provided to validate the effectiveness of the proposed method for the task of compression of high-resolution hyperspectral images.

Fig. 8 indicates that most of the existing methods produce severe spatial–spectral distortion, including the grid-like distortion in JPEG2K, the blocky blurriness in SSCNet, and the bright spot diffusion in HiFiCSE. For 3DCAE, while the overall distribution of the objects is vaguely visible, details such as the contours of buildings and the texture of the narrow roads in the farmland are mostly lost. The proposed MSAHiFiC method yields the clearest visual effect with effective spatial detail resolution. However, it introduces some spectral distortion, such as reduced contrast and, most critically, bare land is turned into green land in the third scene, which could lead to interpretation errors.

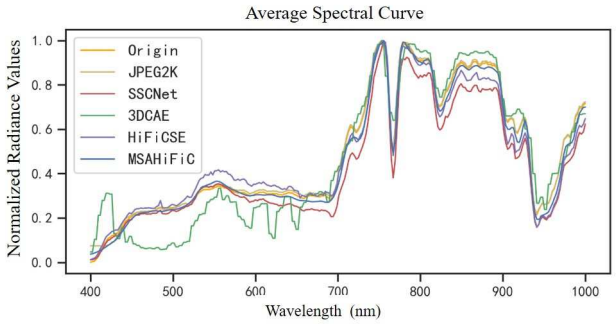
In the reconstructed images of the WHU-Hi-HanChuan dataset shown in Fig. 9, JPEG2K introduces grid-like masking distortion, while 3DCAE results in severe color distortion and the appearance of white borders around the image blocks, indicating a significant loss of image information. In the HiFiCSE-reconstructed images, there is poor alignment between image blocks, and there is considerable mixing of the tree and crop leaf textures, with the spectral information of water bodies and rooftops also nearly lost. SSCNet also suffers from similar texture and spectral distortion, with overall lower image brightness, indicating a severe loss in the radiance peak of the reconstructed images. For the proposed MSAHiFiC method, the texture information of the land cover is well preserved, although the spectral radiance still shows some distortion. As shown in Fig. 11(a), in the normalized average spectral curves of the trees in the WHU-Hi-HanChuan dataset, the spectral band amplitude trends of SSCNet, 3DCAE, and HiFiCSE are noticeably different from those of the original images, indicating that these methods do not effectively capture the spectral interdependencies, resulting in significant spectral distortion. In contrast, JPEG2K and the proposed method show higher spectral fidelity, with only slight shifts at spectral peaks and valleys such as 550, 700, and 800 nm.

For the WHU-Hi-HongHu dataset, as shown in Fig. 10, it is evident that the distortion patterns of the different methods are similar to those observed with the WHU-Hi-HanChuan dataset, with the proposed method still significantly outperforming the others in preserving spatial texture details. Notably, in the large cotton area in the middle of the image, the proposed method accurately reflects the bare soil contained within this region. In contrast, the other methods blur the bare soil in the cotton region. In addition, for the sparsely planted areas in the image, such as Shanghai bok choy,

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(a)



(b)

Fig. 11. Typical reconstruction spectral curves of cotton and trees for the various methods. (a) Reconstructed spectral curves of the tree in the WHU-Hi-HanChuan dataset. (b) Reconstructed spectral curves of cotton in the WHU-Hi-HongHu dataset.

white radish, and Chinese cabbage, the proposed method's reconstructed images tend to lose these small land-cover pixel blocks, making their color representation closer to that of bare soil.

In the average spectral curves for cotton shown in Fig. 11(b), SSCNet and 3DCAE exhibit significant differences from the original image in spectral trends. The other methods can differentiate the spectral peaks, such as those at 550 nm and around 880–900 nm, with HiFiCSE showing more severe spectral distortion, compared to JPEG2K and the proposed method. JPEG2K shows higher fidelity in certain bands, but still shows some differences in radiance due to the varying compression performance across bands. The proposed method's spectral curves are relatively smooth, with minimal radiance fluctuations between the adjacent spectral bands, and the overall trend closely matches that of the original image. Overall, the proposed method better preserves the spatial–spectral details of the land cover.

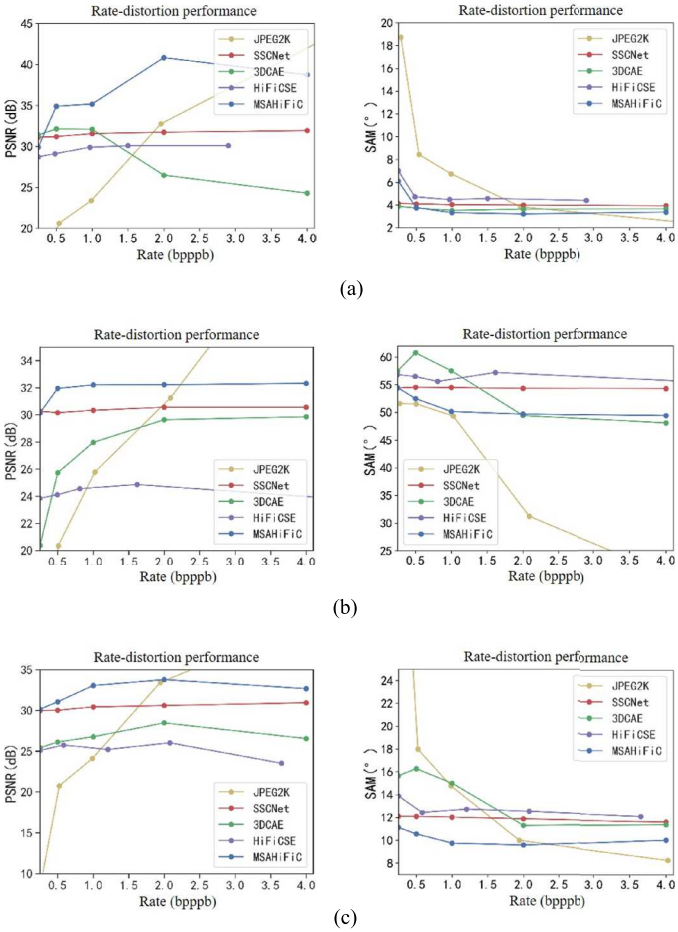
Table V presents the rate-distortion metrics for compressing the WHU-Hi remote sensing image datasets at a rate of 0.5 bpppb (compression ratio of 64). Overall, the proposed method achieves the best performance in most indicators across the WHU-Hi datasets. Notably, due to the varying proportions of the datasets in training, the XiongAn dataset shows a significantly better metric performance compared to the other two datasets. For instance, the PSNR is improved by about 2.2 dB on the XiongAn dataset, compared to the best existing method, whereas the improvement is only 1 dB on the WHU-Hi-HanChuan and WHU-Hi-HongHu datasets. Nevertheless,

TABLE V

RATE-DISTORTION METRIC EVALUATION FOR THE XIONGAN AND WHU-HI DATASETS AT 0.5 BPPPB

Dataset	Metric	JPEG2K	SSCNet	3DCAE	HiFiCSE	MSAHiFiC
XA	SAM	8.443	4.100	3.739	4.725	3.692
	PSNR	20.623	31.204	32.146	29.114	34.379
	MSSIM	0.343	0.784	0.848	0.737	0.901
HC	SAM	51.532	54.546	60.759	56.489	52.505
	PSNR	20.348	30.152	25.726	24.106	31.952
	MSSIM	0.425	0.820	0.392	0.761	0.934
HH	SAM	17.992	12.103	16.275	12.443	10.566
	PSNR	20.749	30.022	26.118	25.756	31.053
	MSSIM	0.458	0.779	0.663	0.691	0.857

Notes: XA: XiongAn, HC: HanChuan, HH: HongHu.



(a)

(b)

(c)

Fig. 12. Rate-distortion curves for the various methods on the WHU-Hi datasets. (a) XiongAn datasets. (b) WHU-Hi-HanChuan datasets. (c) WHU-Hi-HongHu datasets.

the evaluation demonstrates that the proposed method can more effectively preserve the spatial and spectral details of the original images, which is consistent with the visual results discussed earlier.

Table IV lists the rate-distortion metric scores for compression at a rate of 1 bpppb (compression As shown in Fig. 12, the corresponding rate-distortion curves reveal that JPEG2K exhibits an increasingly poor performance with decreasing bit

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rate on the XiongAn dataset. In contrast, the deep learning-based hyperspectral image compression methods effectively extract the key spatial–spectral features, maintaining a relatively high PSNR and low SAM.

Among these deep learning-based methods, HiFiCSE, despite transferring advanced techniques from natural image compression, does not perform well overall. This is mainly because HiFiCSE uses the natural image network structure without modification, which is not suitable for extracting spatial–spectral features from hyperspectral images. In addition, 3DCAE shows a performance trend where the rate-distortion parameters first improve and then deteriorate, which is likely due to the network model being too small and overfitting during training. The SSCNet method demonstrates a relatively stable rate-distortion performance. The proposed method's rate-distortion curve, overall, outperforms those of the other comparison methods, with a higher PSNR and lower spectral angle distance. It is noteworthy that the proposed method exhibits stronger low-dimensional feature representation capabilities, enabling more compact information encoding, effective removal of redundant information, and more efficient compression at low bit rates. The improvement in fidelity diminishes as the bit rate increases, and the performance at 4 bpppb is inferior to that of JPEG2000. Analysis reveals that due to the lack of additional optimization strategies, the proposed model cannot fully utilize the extra bits to restore high-frequency details, resulting in slightly weaker recovery capabilities compared to JPEG2000. This also indicates that deep learning is more suitable for preserving the key features of hyperspectral images under low-bit-rate conditions.

On the WHU-Hi-HanChuan and WHU-Hi-HongHu datasets, the proposed method shows a significantly better PSNR and SAM performance, compared to the other methods, at rates below 2 bpppb (compression ratio of 16). The fidelity improvement due to bit rate is concentrated between 0.25 and 1 bpppb, after which it levels off. The other deep learning-based methods exhibit similar rate-distortion characteristics, while the traditional model compression methods do not show this upper limit. This might be due to the deep learning-based methods using downscaling operations to extract deep features, leading to a loss of texture details and limiting the performance at high bit rates. In addition, the proposed method achieves more accurate bit rate points, compared to the other methods, thanks to the inclusion of bit rate estimation and constraints in the loss function. Overall, the proposed method effectively preserves the key spatial and spectral features across all three datasets, outperforming the other deep learning-based methods and ensuring accurate bit rate estimation and constraint, thus meeting the storage and transmission needs.

As shown in Table VI, the traditional JPEG2K method takes approximately 14 min to encode a single image block at 0.5 bpppb, which is unacceptable for real-time hyperspectral compression needs at the imaging end. Furthermore, since most hyperspectral imaging systems use push-broom scanning, the image input is in strip form, making it unsuitable for the domain transform-based compression methods.

TABLE VI
COMPARISON OF THE VARIOUS METHODS IN HYPERSPECTRAL IMAGE COMPRESSION

Method	JPEG2K	SSCNet	3DCAE	HiFiCSE	MSAHiFiC
MB	—	11.452	1.813	1134.058	208.335
FLOPs	—	9.502	56.027	46.786	38.237
t_E(s)	864.187	0.002	0.003	0.065	0.061

TABLE VII
ABLATION EXPERIMENTAL RESULTS ON THE WHU-HI DATASETS AT 0.5 BPPPB

MASM	VAE	SAM ⇩	PSNR ⇧	MSSIM ⇧
WHU-Hi-HanChuan dataset				
×	×	59.699	26.432	0.523
✓	×	56.012	28.099	0.789
×	✓	56.976	23.806	0.721
✓	✓	52.505	31.952	0.934
WHU-Hi-Honghu dataset				
×	×	14.275	27.088	0.601
✓	×	12.145	28.120	0.735
×	✓	12.963	24.867	0.651
✓	✓	10.566	31.053	0.857

Among the compared deep learning-based methods, the encoding time t_E for the image blocks is currently in the millisecond range, demonstrating the real-time compression performance. In terms of model parameters and computational complexity, 3DCAE has fewer parameters but involves extensive computation, due to the 3-D convolution operations. SSCNet has a suitable parameter amount and complexity for deployment on airborne hyperspectral imaging systems. However, as previously mentioned, its rate-distortion performance is only average, leading to spatial–spectral distortion.

HiFiCSE, while having a high rate-distortion performance, has a high overall complexity and parameter count, making it unsuitable for deployment on resource-constrained edge platforms. The proposed method, although based on the HiFiC framework, includes optimizations to the network structure, effectively reducing the model's parameter count while maintaining a good rate-distortion performance. This suggests its potential for real-time applications in hyperspectral remote sensing, such as on-the-fly collection, compression, and transmission of data. However, it is acknowledged that the proposed MSAHiFiC method still has a relatively high parameter count and complexity, and further optimization techniques, such as model pruning and knowledge distillation, could be considered to enhance its efficiency.

D. Ablation Experiments With the Whu-Hi Datasets

In order to demonstrate the effectiveness of the proposed module, comprehensive ablation experiments have been conducted to evaluate the impact of the MSAM and the VAE structure on the overall performance of the proposed MSAHiFiC model, and the WHU-Hi datasets were used as an example. The ablation results have been provided in Table VII;

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the WHU-Hi-Hanchuan and WHU-Hi-Honghu datasets have been tested, and nearly identical results were observed.

First, the first row (without either structure) clearly yields the worst performance. Second, there is only one module in each of the middle two rows; the results demonstrate that the MSAM module contributes more significantly than the VAE module, further highlighting the importance of spectral channel information in hyperspectral remote sensing data. Finally, the last row represents the proposed MSAHiFiC method, which includes two modules and clearly achieved the best performance, once again proving the effectiveness of the proposed method.

V. Conclusion

To address the limitations of existing deep learning-based hyperspectral image compression methods—specifically, their difficulty in capturing nonlinear spectral dependencies and their crude rate estimation leading to suboptimal rate-distortion performance—this article proposes a hyperspectral compression network incorporating coupled multiscale spectral attention for high-fidelity rate-distortion optimization. The proposed MSAHiFiC method utilizes a super-prior network for rate-distortion analysis of the compressed features, further exploring the spatial–spectral correlations in the compressed features as super-priors, enabling accurate bit rate estimation and constraint optimization. In addition, by introducing the MSAM, the method captures the spectral dependencies across the different bands, ensuring the preservation of critical spectral features during image downscaling. Furthermore, the network's loss function integrates spatial–spectral distortion terms, bit rate constraints, and GAN adversarial loss to enhance the reconstruction fidelity of hyperspectral images. Overall, the proposed method significantly improves spectral fidelity and compression ratio under low-bit-rate conditions.

However, the research still has certain limitations, such as the incomplete optimization of image compression and reconstruction capabilities at high bit rates. In future research, it may be considered to address this issue by introducing more regularization optimization methods into the network.

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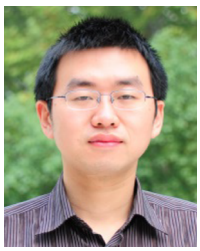
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